

The extracted ore is far from pure and the impurities are removed using physical or chemical methods, such as crushing, grinding, froth flotation or dense media separation. This is followed by a series of chemical reactions to convert the concentrated lithium into the more usable lithium carbonate or hydroxide—a white, powdery substance. Lithium in its purest form, is highly volatile as it ignites on contact with water. It has to be mixed with other metals before it can be used to make products like cathode-active materials that are further used in making lithium-ion batteries. After one more purification step, it is shipped to be used in batteries, ceramics, glass and other products. Based on the experience from global mining projects, De of KPMG reckons it can take up to seven to eight years to bring a mine to full commercial production.

Today, the world's lithium mines are spread across Argentina, Bolivia, Australia and a few other regions in South America. "But the output of those mines is taken to China for final refining. Even though China's lithium reserves are limited, it has managed to dominate the lithium value chain. And 90 per cent of battery-grade lithium carbonate or hydroxide is produced in China," explains Nitin Gupta, CEO & Co-founder of Attero Recycling.

But China's advantage is not a recent phenomenon. "China was the first mover. And now, it has huge processing plants to convert lithium into lithium carbonate and then into cathode activity material. That's where China's strengths lie," says Akshay Singhal, Founder & CEO of nanotech start-up Log9 Materials. He further adds that China has provided various financial incentives to countries rich in lithium resources to corner a major chunk of the global supply lines.



"India's annual lithium-ion battery market is expected to grow to 116 GWh [gigawatt hour] by FY30, from 2.6 GWh in FY21, with EVs accounting for 90 per cent of the market"

PRAHALATHAN IYER
CHIEF GENERAL MANAGER,
RESEARCH & ANALYSIS,
INDIA EXIM BANK



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FOUNDER & CEO,
LOG9 MATERIALS

ENVIRONMENTAL STAKE

Ensuring stable supply lines of the elusive mineral is just one part of the challenge, as mining and refining lithium can also be environmentally damaging. "The mining process can have negative environmental impacts similar to that of other metal mines—such as degradation [of soil], water scarcity, creation of tailing dams and water-flow disruptions—unless companies act responsibly," says Sambitosh Mohapatra, Partner and ESG Leader at PwC India.

Unless sustainable mining practices are followed, lithium extraction would unavoidably lead to increased emissions and also cause soil and water contamination. As per the International Energy Agency's (IEA) estimates, production of a tonne of lithium carbon equivalent (LCE) emits 2.8-9.6 tonnes of CO₂. Moreover, production of one tonne of lithium could require approximately 500,000 gallons of water.

In that backdrop and given India's aim to achieve net-zero emissions by 2070, experts that BT spoke to expressed hope that the government will conduct the mining in a sustainable manner.

WHO'LL MINE, WHERE?

The government now plans to invite bids for the auction of this discovery in the June 2023 quarter and intends to mandate the reserves to be refined domestically only. This could be a stumbling block given that India does not have the expertise or even a lithium mining company. "Lithium mining is very different compared to iron or copper mining, and mining firms in India don't have any experience in that. So, either a company will have to work with some international company that is already in the business, go into a JV, or get into those